

LAB-REPORT → Mobile charger

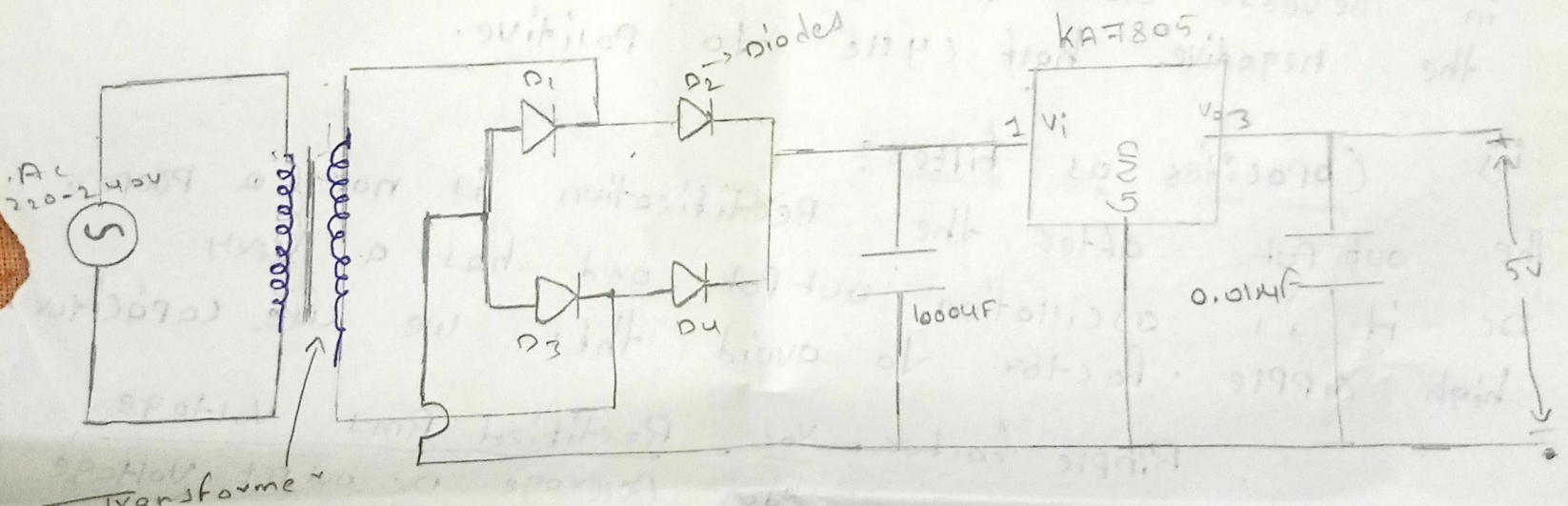
Aim:

The objective of the project is to construct and demonstrate a mobile charger.

→ Components

- 1) Diodes
- 2) 1000μF Capacitor
- 3) 0.01μF Capacitor
- 4) Transformer (230V → 12V)
- 5) 7805 IC Regulator
- 6) USB connector
- 7) Bread ~~board~~ ^{board}

→ Circuit Diagram:



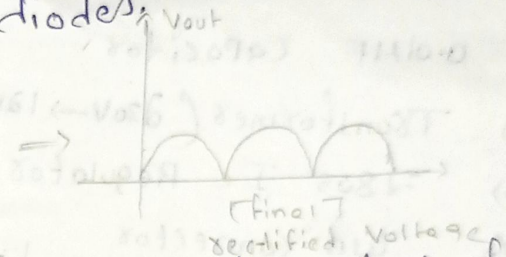
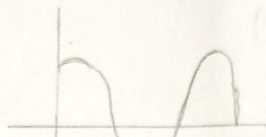
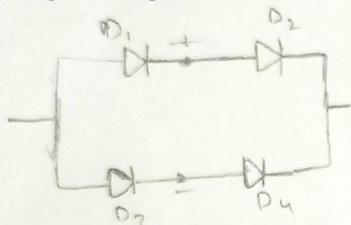
Working procedure of the circuit:

i) Transformer:

To convert 220V AC to 5V DC we need a step-down transformer such that it reduce high voltage.

ii) Diodes:

The process of removing the negative part of the alternate current is called Rectification. It consists the combination of the diodes.

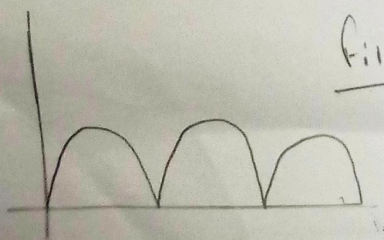


These combination of diodes during the first half of cycle, when voltage flow is in positive direction D_2 and D_3 are in forward bias and D_1 and D_4 are in reverse bias. The bridge wave rectifier converts the negative half cycle to positive.

iii) Capacitor as filter:

The output after the Rectification is not a proper DC. It is oscillation output and has a very high ripple factor. To avoid this we use capacitor.

$$\text{Ripple factor} = \frac{V_{ac}}{V_{dc}} = \frac{\text{Rectified RMS Voltage}}{\text{Average DC output Voltage}}$$



Consider Value of Filter Capacitor:

$$C = \frac{I \times t}{V}$$

Where,

I = maximum output current

C = capacitance

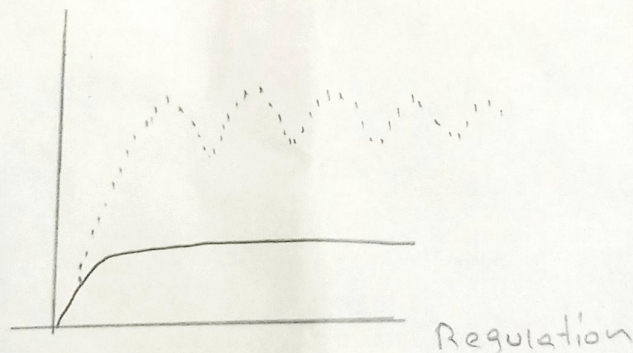
$t = 1/f$; Voltage before rectification had frequency of 50Hz and after rectification it had 100Hz

$$\therefore t = 1/100 \text{ s}^{-1} = 10 \text{ ms}$$

→ 7805 Voltage Regulator

A Voltage Regulator IC 7805 is used to provide a regulated 5V DC. The input voltage should be at least 2-3 Volts more than the output voltage value and the minimum required voltage is 7V.

A 0.01 μF is connected to output in order to eliminate the noise, produced due to transient changes in the voltage.



Precautions

- 1) Be careful while making the circuit as 230V AC mains is being used.
- 2) As the capacitor is bipolar, ensure that the positive and negative terminals are placed properly while making the connections.